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Upper Limb Pain and Dysfunction

MICHAEL BARIA AND EDWARD LASKOWSKI

SUMMARY PEARLS

- The upper extremity is anatomically complex and functionally dynamic, and therefore assessment of the local problem in the context of the kinetic chain is vital.
- The physical examination should be interpreted in light of the patient's demographics, activity, and focused history.
- Although most upper extremity pathology can be managed nonoperatively, special attention should be given to those diagnoses where an expedient surgical referral is necessary, such as acute, full-thickness tears of the rotator cuff, pectoralis major, and distal biceps tendons, as well as scaphoid fracture and scapholunate dissociation.
- Special emphasis should be given to diagnostic entities where a breadth of pathology can impact the upper limb, such as the spectrum of rotator cuff diseases and shoulder instability.

Introduction

"The whole is greater than the sum of its parts." This statement, commonly attributed to Aristotle, is surely manifested in the upper limb. Through the complex interplay of neuromusculoskeletal elements in the upper limb, activities as disparate as playing a piano and throwing a shotput can be achieved. The diversity of functional roles performed by the upper limb leads to a similar diversity in the multitude of injuries that can occur in this anatomic region. Table 1.8 includes photos and additional descriptions of many of the tests described in this chapter.

Upper Limb Physical Examination

Shoulder Special Tests

The anatomic and functional complexity of the shoulder requires that each physical exam maneuver be interpreted in the context of patient history, demographics, and activity/sporting level. The physical exam should be an extension of the medical history and should be performed considering how the response of each maneuver will advance the diagnostic conclusion.

Anterior Apprehension and Relocation Tests

These maneuvers test for anterior glenohumeral joint instability. The patient is placed in the supine position. The examiner abducts the shoulder and flexes the elbow to 90 degrees. The examiner then uses one hand to externally rotate the patient's humerus using the patient's forearm as the lever. At the same time, the examiner's other hand is placed posterior to the patient's proximal humerus and exerts an anteriorly directed force on the humeral head.

The test is positive if the patient indicates a feeling of impending anterior dislocation. To perform the relocation test, the examiner places a hand on the anterior humeral head and exerts a posteriorly directed force. The test is positive if the patient reports relief of apprehension.⁷⁷

Posterior Apprehension Test

This test evaluates posterior glenohumeral joint stability. The patient's affected shoulder is forward flexed to 90 degrees and then maximally internally rotated. A posteriorly directed force is then placed in line with the humerus. If there the humeral head translates posteriorly by at least 50% or the patient feels apprehension, the test is positive.⁷⁷

Sulcus Sign

This test indicates inferior glenohumeral joint instability. The patient is seated or standing with the arm relaxed at the side. The examiner applies a traction force (gripping at either the elbow or forearm). If the patient has inferior laxity, a visible sulcus will be seen at the lateral deltoid, between the humeral head and the acromion.⁷⁷

O'Brien's Test

This test assesses for both acromioclavicular (AC) joint and labral pathology. It consists of two positions. In position 1, the shoulder is flexed forward to 90 degrees with the elbow fully extended. The arm is then adducted 15 degrees, and the shoulder is internally rotated so that the patient's thumb is pointing down. Ensure that the rotation originates from the humerus, not solely the forearm. The patient then resists the examiner's downward force. In position 2, the shoulder is then externally rotated so that the patient's

palm is facing up, and the patient again resists the downward force from the examiner. Pathology of the AC joint causes pain directly over the AC joint in both position 1 and 2. Glenoid labral pathology causes pain and/or clicking/displacement clicking in position 1, which is relieved in position 2.⁹⁵

Horizontal Adduction (Scarf) Test

This test traditionally is used to assess AC joint pathology. The shoulder is passively flexed to 90 degrees and then adducted across the chest (as if the patient is donning a scarf). Pain located at the AC joint suggests AC joint pathology.⁷⁹

Speed's Test

This test evaluates for proximal biceps tendon pathology. The shoulder is forward flexed to 90 degrees with the elbow extended and palm supinated. The examiner applies a downward force at the forearm against the patient's active resistance. Pain in the region of the bicipital groove suggests proximal biceps tendon pathology.⁷⁷

Impingement Testing

Neer-Walsh Maneuver. With the patient's arm at the side, the shoulder is internally rotated. The shoulder is passively forward flexed to 180 degrees while maintaining internal rotation. Pain appreciated near full flexion in the subacromial area indicates rotator cuff involvement (in the acromiohumeral interval), which may include rotator cuff tendon disease.⁷⁷

Hawkins-Kennedy Maneuver. The shoulder and elbow are simultaneously passively flexed to 90 degrees. The examiner then grasps the patient's forearm, stabilizes the scapula, and uses the forearm as to internally rotate the glenohumeral joint. Pain indicates rotator cuff involvement (in the acromiohumeral interval), which may include rotator cuff tendon disease.⁷⁷

Drop Arm Test

The examiner passively abducts the patient's shoulder 90 degrees in the scapular plane (usually about 10–20 degrees of adduction). The patient is asked to slowly lower the arm down. Inability to slowly lower the arm to the side, resulting in pain and a sudden drop of the arm, may indicate a high-grade rotator cuff tear or significant rotator cuff inflammation.⁷⁷

Empty Can/Full Can Test

Traditionally, the empty can test is considered a gold standard physical exam maneuver for evaluation of rotator cuff tears. For this test, the shoulder is abducted to 90 degrees in the plane of the scapula, and the patient is instructed to rotate the thumb down (as if to empty a can). The aim of this rotation is to isolate the rotator cuff by reducing deltoid activation. The examiner then performs manual resistance, and pain plus weakness can indicate a rotator cuff tear. However, this exam maneuver has two major limitations. First, when turning the thumb down, many patients only rotate through the forearm; thus the deltoid activation is never reduced, and supraspinatus is not isolated. Second, if the patient does internally rotate at the shoulder, this causes impingement, increases pain, and may lead to an overdiagnosis of high-grade rotator cuff tears. For this reason, the authors recommend the full can test, in which the same resistance is performed with the patient holding the thumb up (as if holding a full can). Electrophysiologic studies have shown no difference in supraspinatus activation between full and empty can positioning.^{58,62,104}

Additionally, a full can reduces impingement-related pain and allows for a more precise examination of rotator cuff strength.⁴³

Hornblower's Test

The patient flexes and adducts the shoulder with the elbow bent to 90 degrees as if blowing a horn or trumpet. The patient holds this position and resists the examiner's attempt to internally rotate the shoulder. This external rotation force will test both supraspinatus and infraspinatus. Pain and weakness indicate a rotator cuff injury.^{26,49}

Belly Press Test

This test assesses subscapularis injuries. The patient places the fingertips on the umbilicus and elevates both elbows forward. This movement is causing shoulder internal rotation while eliminating pectoralis major involvement. A positive test is present when the patient's elbow is unable to be brought forward.⁴³

Gerber Lift-Off Test

This also tests the subscapularis because it mimics the movements mentioned in the belly press test. The patient places a hand on own lower back and then attempts to push the hand posteriorly, effectively lifting the hand off the back. This is effective in isolating subscapularis, but care should be taken to observe pain inhibition in this position. This will provoke pain with many shoulder conditions and worsen impingement-related symptoms, so all pain should not be attributed to subscapularis pathology.^{26,49}

Elbow Special Tests

Cozen's Test

This evaluates for common extensor tendinopathy (tennis elbow). With the elbow extended and forearm pronated, the patient actively extends the wrist against resistance. Lateral elbow pain suggests pathology at the common extensor tendon.⁷⁵

Mill's Test

This also evaluates for tennis elbow. With a fully extended elbow, the wrist is passively flexed and maximally pronated, causing additional stretch of the common extensor tendon. Pain indicates pathology at the lateral elbow.¹²⁰

Hook Test

This assesses for a distal biceps tendon rupture. With the elbow flexed to 90 degrees and forearm supinated, the patient performs an isometric contraction of the biceps muscle. The examiner then sweeps lateral to medial to assess if the distal biceps tendon is intact. Care should be taken not to confuse an intact lacertus fibrosus with the biceps tendon.⁸⁸

Moving Valgus Stress

This assesses for ulnar collateral ligament (UCL) tears. With the patient seated or standing, the shoulder is abducted to 90 degrees, and the elbow is passively flexed to 120 degrees. The examiner then grabs the wrist and applies a posteriorly directed force to maximally externally rotate the shoulder. The posterior force is maintained, and the elbow is extended. Medial elbow pain between 60 and 120 degrees is a positive test (see Table 36.2, later).⁴²

Ligamentous Instability Test

The examiner flexes the patient's elbow 20 to 30 degrees and stabilizes the patient's arm by placing one hand at the elbow and the other on the distal forearm. Varus and valgus forces are placed across the elbow by the examiner to test the stability of the radial and ulnar collateral ligaments, respectively.⁷⁵

WRIST and Hand Special Tests

Finkelstein Test

This test is used to detect tenosynovitis of the extensor pollicis brevis and abductor pollicis longus tendons (de Quervain tenosynovitis). With the forearm in neutral position, the patient makes a fist with the thumb inside the fingers, and the examiner passively deviates the wrist in an ulnar direction. A positive test result causes pain in the distribution of the affected tendons.⁷⁶

Watson Test

This test assesses scapholunate stability. The patient's wrist begins in an ulnarly deviated position. The examiner places a dorsally directed force against the proximal volar pole of the scaphoid. The examiner then radially deviates the wrist while continuing to place the same force against the scaphoid. A "pop" or subluxation of the scaphoid indicates a positive test result.⁷⁶

Rehabilitation Principles of Upper Limb Injury

The importance of making the correct diagnosis in planning an appropriate treatment program cannot be overemphasized. A complete diagnosis includes whether the injury is acute, chronic, or an acute exacerbation of a chronic injury. Understanding the mechanism of injury is important. Tissues that are overloaded, as well as those directly injured, must be identified. Functional biomechanical deficits such as strength and flexibility imbalances are frequently present. The patient often tries to compensate for the injury by altering movement patterns and using muscle substitutions, which impair effective force funneling in the kinetic chain, reduce functional performance, and contribute to secondary injuries in the kinetic chain at sites distant from the site of pain.

Once an accurate diagnosis has been made with a thorough history, physical examination, and appropriate diagnostic testing, an effective treatment program can be developed. Kibler⁶¹ has proposed three broad stages of rehabilitation: the acute stage, the recovery stage, and the functional stage. The acute stage of rehabilitation focuses on reducing the patient's symptoms and facilitating tissue healing. In specific circumstances, immobilization through splinting or casting may be used during the acute stage of rehabilitation.

Protection, rest, ice, compression, and elevation (PRICE) is frequently used in the acute phase of rehabilitation. Rest should not be absolute. It is important for the patient to maintain cardiovascular fitness, strength, and flexibility during this phase. Core strengthening and aerobic conditioning should be emphasized. The patient should be instructed on appropriate activities that can be performed during the acute stage of rehabilitation that will not aggravate symptoms or be detrimental to tissue healing. For example, if a volleyball player has rotator cuff tendinopathy, activities that could be done include passive glenohumeral joint range of motion (ROM) exercises to maintain flexibility and glenohumeral joint health, riding a stationary bicycle for cardiovascular fitness, and performing

scapular stabilizing exercises in preparation for more advanced rehabilitative exercises of the shoulder. Kinetic chain deficits should be identified and "weak links" treated during the acute rehabilitation stage.⁵

During the acute phase of rehabilitation, cryotherapy can be used for acute injuries to decrease pain, inflammation, muscle guarding, and edema.^{50,57} Heat increases blood flow, reduces muscle spasm, reduces pain, and can be used in the acute phase of rehabilitation of chronic injuries.⁵⁷ High-frequency electrical stimulation may also be used during the acute phase of rehabilitation to reduce muscle guarding and increase local circulation.¹³²

Opioid and nonopioid analgesics may be required for pain control during the acute phase of rehabilitation. Nonsteroidal antiinflammatory drugs (NSAIDs) are often used for their analgesic and antiinflammatory properties. Randomized placebo-controlled trials have demonstrated reduced pain, edema, tenderness, and a faster return to activity in NSAID-treated athletes than in those treated with placebo.^{6,129} It is important to remember, however, that NSAIDs are not entirely benign and can cause significant gastrointestinal, renal, cardiovascular, hematologic, dermatologic, and neurologic side effects.^{40,124,221} Because of these concerns, NSAIDs should be used judiciously.

Oral and injected corticosteroids have also been used for pain control and reduction of inflammation during the acute phase of rehabilitation. Because of the possibility of significant systemic and localized effects, the use of corticosteroids should be limited to select cases.^{32,112} Potential complications include suppression of the avascular necrosis, skin depigmentation, fat atrophy, infection, or tendon rupture.²⁴

The patient can advance to the recovery phase of rehabilitation when pain has been adequately controlled and tissue healing has occurred.⁶¹ This is indicated by full pain-free ROM and the ability to participate in strengthening exercises for the injured limb. The emphasis of the recovery phase of rehabilitation involves the restoration of flexibility, strength, and proprioception in the affected limb. Strength and flexibility imbalances, maladaptive movement patterns, and muscle substitutions should be corrected in this phase of rehabilitation. Open kinetic chain exercises can be beneficial when strength imbalances are being corrected, whereas closed chain exercises are frequently used to provide joint stabilization through muscle cocontraction. Cardiovascular and general fitness should be maintained with progression to functional activities toward the end of this phase. Blood flow restriction training can be used to provide a strength stimulus for healing tissue at a low load and to start strength progression at an earlier stage.^{52a}

The patient can begin the functional stage of rehabilitation when the injured limb has regained 75% to 80% of normal strength compared with the uninjured limb and when there are no strength and flexibility imbalances.⁶¹ The patient's rehabilitation must continue to address maladaptive movement patterns and muscle substitutions, and full strength should be obtained during this stage. Functional activities should be incorporated into the rehabilitation program with a vocational/avocational or sports-specific progression that eventually leads to a return to normal activities.

Musculoskeletal Problems of the Upper Limb

Conditions of the Shoulder

Acromioclavicular Joint Sprains

AC joint sprains account for 9% of all shoulder injuries and are most frequent in males in their third decade of life.⁷⁹ Most injuries

occur as a result of direct trauma from a fall or blow to the acromion. Physical examination demonstrates point tenderness, a positive horizontal adduction (Scarf sign) test, and pain in both positions 1 and 2 of O'Brien's test.

AC joint sprains are classified into six types (Fig. 36.1).

Type 1: AC ligament sprain only with normal radiographs.

Type 2: AC ligament disruption but with intact coracoclavicular ligaments. Radiographs demonstrate clavicular elevation relative to the acromion but less than 25% displacement.

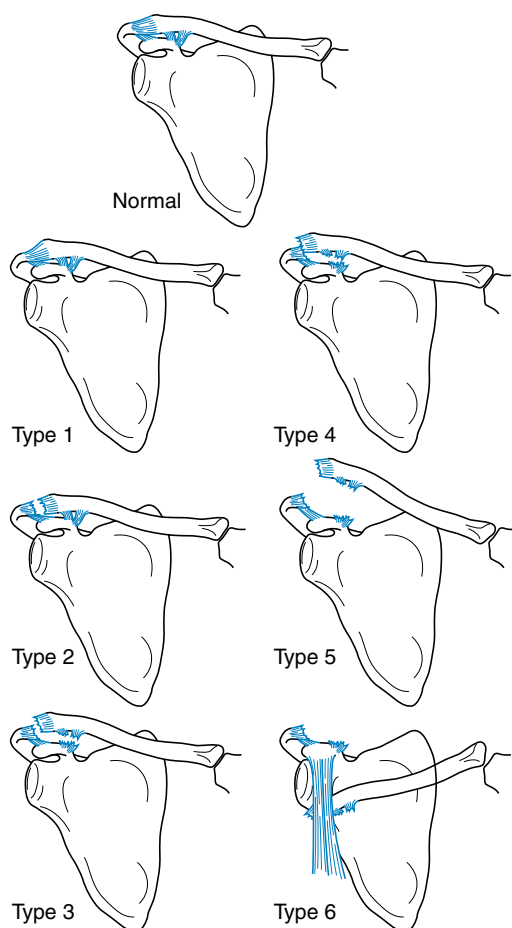
Type 3: Disruption of the AC and coracoclavicular ligaments, but the deltotracheal fascia remains intact. Radiographs reveal a 25% to 100% increase in the coracoclavicular interspace relative to the normal shoulder.

Type 4: Disruption of the coracoclavicular and AC ligaments, with posterior displacement of the distal clavicle into the trapezius muscle.

Type 5: Disruption of coracoclavicular and AC ligaments and clavicular displacement through the deltotracheal fascia. This results in an increase in the coracoclavicular interspace to greater than 100% of the normal shoulder.

Type 6: Disruption of the coracoclavicular and AC ligaments with inferior clavicular displacement below the coracoid process.

Radiographic evaluation of the AC joint should include anteroposterior and lateral views of the AC joint and a Zanca view (anteroposterior projection with 15 degrees cephalic tilt). Though commonly obtained in the past, stress views do not provide additional clinically useful information.¹⁰⁸



• Fig. 36.1 The Rockwood classification of acromioclavicular joint sprains.

Type 1, 2, and 3 AC joint sprains are usually treated nonoperatively. A brief period of sling immobilization may be used to reduce pain. Indications for surgical intervention for type 3 sprains include persistent pain and AC joint instability despite conservative care or unsatisfactory cosmetic results. Some authors advocate surgical treatment of type 3 sprains in athletes who participate in sports that place a high demand on the upper limbs,^{20,67,103} but the current literature favors nonoperative treatment for type 3 sprains.¹²¹ Type 4, 5, and 6 sprains require surgical consultation.⁷⁹

Biceps Pathology

The biceps muscle-tendon unit crosses both the shoulder and elbow joint. It originates from the superior glenoid-labral complex and inserts at the radial tuberosity. It performs a variety of functions, including contributing to the stability and flexion of the glenohumeral joint, flexion of the elbow, and supination of the forearm. Because the biceps muscle tendon unit crosses two joints and confers both strength and stability to the upper limb, it is prone to injuries along its entirety.²⁸

Proximal Biceps Tendinopathy. The tendon of the long head of the biceps is part of the rotator interval and is critical for normal shoulder stability and biomechanics. The tendon can be damaged by the same impingement that causes chronic degeneration of the rotator cuff, which is why proximal biceps tendinopathy often coexists with rotator cuff tendinopathy. The proximal biceps tendon is also commonly injured by repetitive traction forces, such as overhead throwing. Patients with chronic proximal biceps tendinopathy often present with atraumatic anterior shoulder pain that is exacerbated by reaching overhead and throwing. They will be tender in the bicipital groove and have pain reproduced with Speed test. Because the proximal tendon originates from the superior labrum, O'Brien test may also provoke shoulder pain. However, pain originating from the biceps tendon localizes to the anterior shoulder (rather than deeper in the joint) and is not usually associated with mechanical symptoms. Treatment for proximal biceps tendinopathy includes optimization of shoulder mechanics, restoration of normal scapular motion to relieve subacromial impingement, and shoulder proprioceptive training. Biceps tendon sheath injections may be a useful adjunct. Patients with refractory symptoms may benefit from surgical tenotomy/tenodesis.^{19,82}

Long Head of Biceps Tendon Rupture. The proximal biceps tendon may also be injured by an acute rupture. This can occur with or without trauma. In the absence of trauma, the patient is typically older and has a history of chronic rotator cuff disease. Trauma, such as a fall or catching a heavy object, can rupture the proximal biceps tendon. Patients with a proximal biceps tendon rupture will present with a classic Popeye deformity, in which the muscle belly will retract toward the elbow and result in a visible asymmetry. Most proximal biceps tendon ruptures are treated nonoperatively. Functionally, a proximal rupture results in only 10% loss of elbow flexion strength (because of the greater contributions from brachialis and brachioradialis). If the patient must lift heavier loads for work or sport, a tenodesis may be performed to optimize function.²⁸

Distal Biceps Tendon Rupture. This usually result from sudden eccentric load (i.e., catching a heavy object while pulling up in elbow flexion). Heavy deadlifting is another common sporting event implicated in this injury because the tendon is fully extended and then placed under significant eccentric force during the initial pulling phase of the lift. Patients will describe sudden deep pain,

usually with a “pop” deep in the elbow. Ecchymosis and a visible deformity are observable (as the muscle retracts proximally). Distal biceps tendon rupture can be diagnosed on physical examination by a positive Hook test. In some cases, a fully ruptured distal tendon can still feel intact because it is held in place by the lacertus fibrosus. In the setting of a concerning injury, additional testing should include resisted elbow flexion and resisted forearm supination. Pain and weakness with these movements should raise the suspicion for a high-grade distal tendon injury and prompt additional evaluation with MRI. If a distal rupture is confirmed, a timely surgical referral is key because fixation is required for optimal strength, function, and cosmesis. Delays in surgical care are often complicated by significant fibrosis.⁵⁵

Rotator Cuff Tendon Disease

The rotator cuff (comprised of the supraspinatus, infraspinatus, teres minor, and subscapularis) is impacted by a large spectrum of pathology. The primary pathologies include (1) tendinopathy without significant tearing, (2) partial tears, (3) chronic full-thickness tears, and (4) acute, traumatic full-thickness tears. Though the nuances of each are unique, they are most easily categorized according to their primary treatment method. As such, we will discuss them in the following categories: nonsurgical (tendinosis, partial tears, and chronic full-thickness tears) and surgical (acute full-thickness tears).

Primary Treatment: Nonsurgical. Tendinopathy, partial tears, and chronic full-thickness tears are best treated first with a nonsurgical, rehabilitation-based treatment plan. Surgery should only be considered for these ailments after patients fail comprehensive nonoperative treatment.

The etiology of these ailments is multifactorial, and contributing factors include overuse, poor biomechanics, microtrauma, minor injuries, and a genetic predisposition. These conditions are often chronic because of the poor healing capacity of the rotator cuff. Microvascular studies demonstrated a hypovascular zone on the articular surface of the leading edge of the supraspinatus tendon.^{74,105} This region has been implicated in rotator cuff degeneration and correlates with the higher incidence of articular-sided, partial-thickness tears in the anterior aspect of the supraspinatus tendon.

Role of Impingement in the Development of Chronic Rotator Cuff Dysfunction. The impingement of the rotator cuff can be external or internal. External impingement is more commonly referred to as subacromial impingement. This occurs when the arm is raised overhead, and the greater tuberosity enters the subacromial space. If the patient has poor scapulothoracic mechanics, the dyskinetic movement does not allow for normal clearance of the subacromial space. The rotator cuff then becomes impinged, and chronic frictional damage occurs. This impingement must be addressed during rehabilitation to prevent ongoing injury.⁷³

Internal impingement is less common. It occurs overhead in athletes when the arm is abducted and externally rotated. If there are proprioceptive deficiencies or subtle increases in shoulder laxity, this can lead to contact and friction between the external rotators and the posterior glenoid. With increased laxity, the humeral head translates anteriorly, which accentuates the impingement.^{7,52}

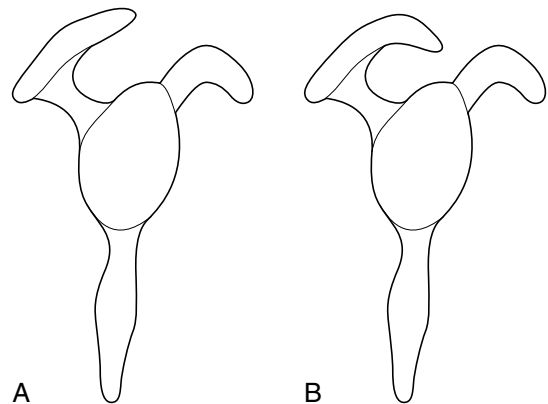
Clinical Presentation. Shoulder impingement patients will present a variety of historical scenarios: They may be entirely atraumatic, with gradual onset of symptoms; they may have an injury mechanism, such as catching a heavy falling object; or there may be a new (or more intense) activity, such as a vigorous exercise routine, or more consistent or higher velocity overhead throwing.⁵²

Symptomatically, patients note anterolateral shoulder and lateral arm pain exacerbated by overhead activity, reaching behind the back, lifting, and lateral side lying on the affected side. Symptoms such as stiffness, weakness, and catching/clicking may be present.⁵²

On examination, observe for scapular dyskinesis (an upward, outwards rotation of the scapula with overhead arm movement caused by scapular instability) because this may be the foundational cause of impingement. Positive impingement testing in the absence of trauma is also useful for establishing impingement as a contributing factor. Examiners should also screen for significant ROM restriction. Rotator cuff dysfunction can lead to mild ROM deficits, usually from pain avoidance and disuse, and most commonly involving decreased internal rotation. However, inability to abduct beyond 90 degrees and restricted external rotation indicate either adhesive capsulitis or advanced osteoarthritis (OA) rather than rotator cuff disease. Rotator cuff strength testing consists of full can, drop arm, hornblower, belly press, and Gerber lift-off test. Weakness with one or more of these should raise suspicion of significant tendinopathy or a higher grade tear. Specifically, weakness with full can, drop arm, and hornblower tests usually indicates injury to the supraspinatus. Weakness with resisted external rotation may indicate injury to supraspinatus or infraspinatus. A positive belly press and/or Gerber lift-off test indicates subscapularis injury.

When performing these rotator cuff special tests, care should be taken to distinguish between pain and pain-limited weakness vs. weakness caused by high-grade injury. Pain and pain-limited weakness typically indicate tendinosis or a partial tear. Significant weakness in the absence of a major injury may indicate a chronic, degenerative, full-thickness rotator cuff tear.⁶⁰

Radiographic evaluation of patients with suspected rotator cuff pathology should include anteroposterior, supraspinatus outlet, and axillary radiographs. Anteroposterior radiographs should be performed in the neutral, external, and internal rotation positions to adequately visualize the glenohumeral joint and the greater and lesser tuberosities. Full-thickness rotator cuff tears may demonstrate a narrow acromiohumeral distance of less than 7 mm and sclerosis on the undersurface of the acromion. This finding occurs because, in the absence of the supraspinatus, the deltoid contracts unopposed, causing superior humeral head migration. This process takes time to develop and should not be relied upon to assess for an acute full-thickness injury. The supraspinatus outlet view allows for categorization of the acromion type (Fig. 36.2) and will reveal AC joint osteophytes. Ultrasound and MRI have higher



• **Fig. 36.2** (A) Type 1 acromion. (B) Type 3 acromion.

levels of sensitivity and specificity for rotator cuff pathology than radiographs.^{13,53}

Treatment Approach. Because tendinosis leads to partial tendon tears, it is self-evident why these entities are grouped together for treatment. It may initially seem counterintuitive that a chronic full-thickness tear would undergo a similar treatment approach. However, a chronic tear of significant magnitude of one or multiple rotator cuff tendons is inoperable because the retracted tissue undergoes fatty infiltration and fibrosis. Without a possible direct repair, a restorative surgery does not exist.^{65,128} Consequently, the same treatment approach is used for partial tears or chronic full-thickness tears of the rotator cuff.

Rehabilitation-based care should focus on pain reduction, correction of biomechanical abnormalities, and restoration of function. In the acute rehabilitation stage, emphasis should be on scapular stabilizers, rather than the rotator cuff itself, and establishment of a balanced shoulder force couple. Specifically, strengthening muscles that stabilize (e.g., the serratus anterior), retract, and depress the scapula (e.g., rhomboids and inferior trapezius) and stretching muscles that protract and elevate the scapula (e.g., pectoralis minor, upper trapezius, and levator scapulae) can help to mechanically reduce impingement by establishing anterior-posterior muscle balance and reducing forward shoulder protraction. Restoring normal scapulothoracic mechanics will alleviate functional impingement and reduce mechanical strain of the rotator cuff. Next, direct strengthening of the rotator cuff should be implemented. Closed chain exercises can help to promote stability and proprioception, while open chain exercises can be used to correct strength imbalances, such as disproportionate weakness of the external rotators. If pain persists or interferes with therapeutic exercise, work, or sleep, a subacromial corticosteroid injection may be helpful.^{48,83,89}

Primary Treatment: Surgical

In contrast to the rotator cuff pathologies discussed earlier, acute traumatic full-thickness tears should be referred for surgical consultation to optimize their repair potential. Acute, full-thickness injuries are distinct and more urgent because, if left unrepaired, the patient will often end up with significant disability of the upper limb. Because of the unopposed deltoid contraction, the humeral head migrates superiorly, obliterating the normal subacromial space and resulting in rotator cuff arthropathy, which is a unique pattern of advanced degenerative joint disease. This can be prevented by expedited rotator cuff repair.¹²⁸

Physicians should maintain a high index of suspicion for an acute, full-thickness rotator cuff tear in patients who sustained a significant injury, such as a fall on an outstretched hand (FOOSH) or suddenly catching a heavy object. These patients will experience severe, sudden pain and significant functional decline, often resulting in inability to use the affected shoulder. These patients should be promptly evaluated with an MRI and surgical consultation. Even a delay as little as 6 weeks will result in fatty infiltration of the muscle and decrease the chances of a successful surgical outcome.

Glenohumeral Joint Instability

Glenohumeral joint stability is provided by static and dynamic stabilizers. The static stabilizers include the inferior, middle, and superior glenohumeral ligaments, as well as the articular surface and labrum.¹ The dynamic stabilizers include the rotator cuff and the long head of biceps.⁶⁸ The rotator cuff contributes to dynamic stability through a number of mechanisms. Concave compression, first described by Lippitt et al.,⁷¹ refers to the compressive forces

placed on the glenohumeral joint during rotator cuff activation. These forces center the humeral head into the glenoid, helping resist joint translation. Because the glenohumeral ligaments are lax in the midranges of glenohumeral joint motion, coordinated rotator cuff contraction and concave compression are vital mechanisms for stability.⁶⁸ The distal rotator cuff tendon insertions are interwoven with the joint capsule. This then allows for capsular tightening during rotator cuff activation, further accentuating the ability of the cuff-capsule complex to improve joint stability.⁶⁸

When these stabilizing structures are compromised, the resulting instability is characterized in two large categories: traumatic/unidirectional and atraumatic/multidirectional.

Traumatic Instability

Traumatic joint dislocations occur either in the anterior or the posterior direction. Classically, a traumatic, anteroinferior dislocation occurs with high-velocity energy to the upper limb when the shoulder is abducted, extended, and externally rotated. In this position, the inferior glenohumeral ligament is maximally stretched, and the supramaximal force results in ligament tearing and joint dislocation. An example of this injury is when a defensive football player is attempting to make a tackle but is out of position and uses the arm only to stop the runner. The clinical presentation is obvious with pain, and the patient is usually able to self-diagnose. On examination, there is visible deformity, with the humeral head sitting anterior. This may give the appearance of shoulder swelling. First, a neurovascular examination should be performed and documented to ensure that a limb-threatening injury is not present. Palpation of the proximal humerus should also be performed to rule out a humeral fracture (especially before an attempt at reduction). If the examination confirms a simple joint dislocation, one or two attempts at reduction should be made. A variety of reduction techniques have been described.²⁹ They all attempt to relax the shoulder, minimize muscle spasm, and create enough axial traction that will allow the joint to reduce. This should always be done gently and slowly to avoid an iatrogenic injury, which has been documented in more forceful reduction attempts. If reduction attempts fail, the patient should be sent to the emergency room for consideration of reduction under sedation. In some cases, patients may present to clinic days after the dislocation and reduction. In this scenario, they will have a positive apprehension and relocation test. Postreduction radiographs should be obtained to document a successful reduction as well as evaluate for bony injury. Radiographs may show a Hill-Sachs lesion (posterolateral humeral head depression from direct impaction) or a bony Bankart lesion (anteroinferior glenoid fracture). MRI is indicated for a patient with a history of multiple dislocations, presence of bony abnormality on x-ray, or when the clinical presentation suggests a concomitant high-grade rotator cuff tear.⁴

Treatment for a first-time traumatic dislocation that is uncomplicated (without significant bony pathology or rotator cuff injury) is generally a rehabilitation-based approach. Acutely, the focus is on pain reduction, a brief period of immobilization (≤ 7 days), and progressive proprioceptive and strengthening in safe zones. Early strength should consist primarily of pain-free closed chain movements to foster joint stability and emphasize muscular cocontraction to provide dynamic stability. As the patient recovers, open chain exercises that further challenge proprioception can be integrated. Surgical intervention is considered for patients who have multiple dislocations or who have bony/soft tissue injuries that significantly increase the risk of recurrent

instability (such as large Hill-Sachs, bony Bankart, or humeral avulsions of the glenohumeral ligaments [HAGL] lesions). Epidemiologically, younger patients, especially those under age 19 years, are at significant risk for repeat dislocations.^{4,46}

Posterior acute dislocations may occur after seizure or electrocution. Postdislocation sequelae include a posterior glenoid labrum tear (a reverse Bankart lesion) or a reverse Hill-Sachs lesion, an impaction fracture of the anterior humeral head.^{9,105} Seizure-related posterior instability can become chronically unstable if the patient has frequent seizures. The combination of forceful muscle contraction plus recurrent soft tissue and bony damage can lower the threshold for each subsequent dislocation. These patients may benefit from surgical stabilization, but long-term stability can still be compromised if the underlying seizure disorder is not well controlled.¹⁰²

Atraumatic Instability

The other distinct category of instability is atraumatic and multidirectional. This typically occurs in patients with hypermobility that impacts many joints. Patients often complain of pain that is difficult to localize, and they can voluntarily sublux the joint (i.e., cause partial translation that is palpable to the patient and physician without full dislocation). They demonstrate laxity in all directions, and this often impacts both shoulders. Hallmark findings on the physical examination include visible voluntary subluxation, a high Beighton (assessment screen for hypermobility) score, and a positive sulcus sign. These patients should be managed with a rehabilitation-based approach. The patient should be educated to not voluntarily sublux the joint; cognitive behavioral therapy may be helpful to aid in diminishing this habit. Strengthening should occur in safe planes and ROMs that do not aggravate laxity. Finally, improving proprioception will help to further improve joint symptoms. Because these patients have increased laxity in connective tissue, surgical correction (often done through capsular plication) often has poor outcomes.¹¹⁴

The physical examination of an acute shoulder dislocation depends on the clinical setting. If an acute dislocation is witnessed, as mentioned, a focused neurovascular exam should be performed to rule out a more worrisome injury. Additionally, palpation and ROM should be done to evaluate for a fracture that would preclude reduction in the field. If the patient is seen in the clinic postreduction, the examination should focus on evaluating for clinical instability with apprehension testing and special tests for labrum and/or rotator cuff injuries.

The most common radiographic evaluation includes the anteroposterior (AP), axillary, and scapular Y views.¹⁴ The AP view visualizes osseous structures.¹¹⁵ The AP view can be augmented with internal rotation, which evaluates for a Hill-Sachs defect.¹¹⁵ The scapular Y and axillary views assess for active and reduced dislocations as well as glenoid fractures.¹¹⁵ Other specialized views include the Garth view and the West Point view, both of which are useful in the detection of Bankart fractures. The Stryker Notch view can be used for evaluation of Hill-Sachs defects and stress views for the documentation of the degree of glenohumeral joint instability (though the latter is not routinely necessary).¹¹⁵ MRI provides the best assessment for soft tissue injury, including the labrum, rotator cuff, and HAGL lesions, all of which may require surgical intervention.

Overall Treatment

The treatment options for glenohumeral joint instability and dislocation include nonoperative and operative approaches.

After initial glenohumeral joint subluxation episodes and in patients with multidirectional instability or unidirectional posterior or inferior instability, a comprehensive rehabilitation program that addresses kinetic chain deficits, scapulothoracic mechanics, shoulder girdle strength, flexibility, and neuromuscular control is appropriate. Operative intervention should be considered only in those patients who have failed to improve after a comprehensive nonoperative treatment program.¹⁰⁷

For patients with glenohumeral joint instability, the strengthening program should begin with closed chain exercises that promote cocontraction of the glenohumeral joint–stabilizing musculature. The patient can eventually progress to functional open chain exercises as stability is achieved. Proprioceptive closed and open chain exercises are also important to establish coordinated firing of the stabilizing shoulder girdle musculature and to attain neuromuscular engrams that respond appropriately to a dynamically changing environment.

For patients who have sustained a first-time traumatic anterior glenohumeral joint dislocation, the decision between a trial of nonoperative treatment vs. immediate surgical stabilization is more controversial. In the older, less active patient, nonoperative management is frequently successful.³ In the younger, more active patient involved in contact sports, studies have shown a very high repeat dislocation rate in those treated nonoperatively compared with those receiving early operative intervention.^{10,12,63,122}

Glenoid labral tears. The glenoid labrum is a static stabilizer of the glenohumeral joint. Given the shallow conformation of the glenoid, the labrum provides increased contact surface and stability for the humeral head. The labrum may be injured because of repetitive movements/overuse, a traumatic injury, or may undergo degenerative changes.

Overuse Labral Pathology. Classically, overhead throwing athletes can injure the labrum because of the repeated forces through the joint. The late cocking and early acceleration phases of throwing lead to increased torsional force from the biceps anchor on the labrum.¹⁵ This is referred to as the “peel-back” mechanism, and this chronic torsion can lead to labral injuries such as tear of the superior labrum from anterior to posterior.³³ Additionally, the follow-through phase increases traction forces from the biceps to the labrum. Athletes with labral tears may feel deep shoulder pain with or without mechanical symptoms (clicking or shifting). They may also complain of loss of control over their throws, or a sensation of “dead arm,” which manifests as arm heaviness or fatigue after throwing. Dead arm may be confused with cervical spine injuries, such as a stinger, because of the pain referral pattern. On examination, patients may demonstrate a loss of glenohumeral ROM (glenohumeral internal rotation deficit [GIRD]) because of a posterior capsular contracture. They also may have a positive O’Brien, crank, and/or dynamic labral shear test. Patients should undergo comprehensive rehabilitation, including throwing mechanics analysis, optimization of hip, spine, and shoulder mobility (including addressing their GIRD), and proper kinetic chain integration. If patients fail to improve with rehabilitation, they may benefit from surgical consultation.⁵¹

Posterior labral tears can also be the result of chronic, posteriorly directed forces. This is most often seen in athletes such as football linemen. As they engage their block with extended arms, the chronic posterior force causes degenerative labral damage. This results in subtle posterior translation during activity. Patients with posterior labral tears may complain of posterior pain when they sustain a posterior axial force. On examination, they may have pain with the posterior apprehension test. Rehabilitation of

these injuries should focus on the optimization of sport-specific technique, proper balancing of the shoulder's force couple, and potentially bracing that prevents excessive adduction of the shoulder.¹¹⁸

Traumatic Labral Tears. These are typically the result of shoulder dislocations. As the humeral head dislocates, it disrupts the labral tissue. These injuries are referred to as a Bankart lesion (after an anterior shoulder dislocation) and reverse Bankart lesion (after a posterior dislocation). Labral injuries increase the risk of a first-time dislocation developing chronic instability. The assessment of these injuries occurs in the context and timing of the shoulder dislocation evaluation. If patients present far after their dislocation, the examination may have positive apprehension testing, as well as positive O'Brien, crank, and dynamic labral shear tests. Treatment of these concomitant injuries is detailed in the shoulder instability section.

Degenerative Labral Tears. These are common in an aging population. They are associated with rotator cuff dysfunction. These are often found incidentally on MRIs performed for another reason. Degenerative labral tears are typically asymptomatic. In the absence of a proper mechanism of injury or objective instability, these findings do not warrant surgery and respond well to rehabilitation focused on a balanced shoulder force couple.

Adhesive Capsulitis

Adhesive capsulitis, or "frozen shoulder," is characterized by a painful, progressively stiff shoulder in all motion planes with normal radiographs (which rule out OA).^{41,91} Adhesive capsulitis occurs in approximately 2% to 5% of the population, is more common in females, and is most frequently seen between ages 40 and 60 years.^{18,23} It is frequently associated with diabetes mellitus, systemic inflammatory disorders, prolonged immobilization (in the setting of medical events such as a stroke or prolonged surgical positioning), and thyroid disease. Histopathology is characterized by synovitis, fibroblastic proliferation, and the formation of nodular bands.¹⁸

Adhesive capsulitis progresses in stages, of which there are three or four, depending on the classification system being referenced (Table 36.1).⁴⁵ Regardless of nomenclature, recognition of the clinical presentation is most important for optimal treatment. Early in the disease course (approximately the first 1–3 months), patients present with painful shoulder movements but with

preserved ROM. Because the hallmark stiffness is absent, this is easy to misinterpret as rotator cuff pathology. Therefore any patient early in symptom presentation who has risk factors as noted should be educated on symptom progression and to return to clinic if stiffness develops. Next, the shoulder progresses into the "freezing stage." This occurs between 3 and 9 months, is associated with persistent and worsening pain, and a significant amount of ROM is lost. Often, patients lose the ability to reach overhead and can no longer perform daily self-care, such as dressing or washing their hair. During the frozen stage, symptoms usually have been present for 9 to 15 months, pain begins to recede (as the inflammatory process resolves), but the stiffness persists. Finally, with the "thawing" stage, stiffness improves for up to 2 to 3 years after initial symptom onset.

The physical and radiographic evaluation of these patients is straightforward. Given the appropriate history, patients will demonstrate a progressive loss of ROM in all planes, with inability to abduct the shoulder beyond 90 degrees, along with a significant loss of external rotation. External rotation is the first movement to be restricted, so this should be an evaluation focus of any early presentation. The radiographic evaluation is often normal, which rules out OA as a cause of pain and stiffness. The progression of stiffness in adhesive capsulitis is also more precipitous than is typically observed with OA, which is far more gradual. Advanced imaging is usually unnecessary, but MRI will demonstrate significant capsular and synovial thickening and inflammation.

During the initial and freezing stages, treatment consists of deep heat modalities, analgesics, and efforts to restore ROM. Pain severity may interfere with ROM exercises; therefore up to three intraarticular corticosteroid injections can be used to treat inflammation and pain, facilitate rehabilitation, and hasten the time to resolution.^{45,117} Once adequate pain control is achieved, joint mobilization along with passive and active assisted ROM exercises should be initiated. As the patient's symptoms improve, active ROM activities can be added, along with open chain and proprioceptive exercises. Integration of active ROM should be done as soon as possible to correct any maladaptive movement patterns that have developed.

If the patient presents later in the frozen stage, steroid injections do not have the same potential benefit because fibrosis has already developed. A steroid injection in this phase still may help with pain, but it does not seem to change the disease course. Some studies indicate that steroid injections given early in the course of the condition can hasten recovery of ROM.⁹⁰

Most patients will have restoration of normal function over a 12- to 14-month period.^{44,86} In patients who do not improve after 6 months of nonoperative treatment, other treatments aimed at resolving fibrotic tissue can be employed. These include capsular hydrodistention, manipulation under anesthesia, and arthroscopic lysis of adhesions and capsular release.^{17,18,31,36,57,63,92,97,99,132} Finally, a growing body of literature supports the use of platelet-rich plasma (PRP) injections for long-term symptom improvement (compared with steroid and physical therapy).¹⁰⁶

Pectoralis Major Ruptures

The pectoralis major muscle is responsible for arm flexion, adduction, and medial rotation. It has distinct sternal and clavicular heads that, as they transition at their myotendinous junction, rotate on each other as they insert on the lateral aspect of the bicipital groove. Injuries to the pectoralis major are rare but must be recognized early because of the potential for significant impairment.

TABLE 36.1 Stages of Adhesive Capsulitis

Stage	Symptom Duration (months)	Signs and Symptoms
1	1–3	Painful shoulder movement, minimal restriction in motion
2	3–9	Painful shoulder movement, progressive loss of glenohumeral joint motion
3	9–15	Reduced pain with shoulder movement, severely restricted glenohumeral joint motion
4	15–24	Minimal pain, progressive normalization of glenohumeral joint motion

Pectoralis major injuries usually occur at the distal tendon because of a large eccentric force. Classically, this occurs during bench press. The patient will feel pain and a pop. Extensive ecchymosis develops in the subsequent hours along with significant muscle deformity. On examination, the ecchymosis is prominent, and there will be a loss of the normal deltoid-pectoral contour. The pectoralis contour will essentially be vertical, instead of oriented toward the biceps groove, because the muscle has retracted. There will be pain and weakness with manual muscle testing of the pectoralis major muscle.^{111,133}

The diagnosis can be made clinically without further imaging. MRI may be obtained for confirmation and surgical planning. The patient should be promptly referred to an orthopedic surgeon because delayed repair will be complicated by advanced scarring.

Conditions of the Elbow

Lateral Epicondylitis (or Epicondylitis)

Lateral epicondylitis is a common tendinopathy of the elbow and is frequently referred to as tennis elbow. This malady dates back to 1883, when it was described as an injury resulting from lawn tennis,²⁷ and it is reported to occur in up to 50% of tennis players.⁹⁴ Any activity that places repetitive stress on the lateral forearm musculature, however, can cause this condition. It is more common in patients older than 35 years of age and peaks in those between ages 40 and 50 years.^{56,93} It is more common in male than in female tennis players, but does not display a sex predilection in the general population.²⁵

The hallmark pathologic change is angiofibroblastic dysplasia, which represents a degenerative (not inflammatory) condition.^{93,108} This primarily impacts the extensor carpi radialis brevis, but also involves the extensor digitorum communis origin in 30% of cases.⁹³ The origins of the extensor carpi radialis longus or extensor carpi ulnaris are rarely involved.

Patients report gradual symptom onset, which is provoked after specific gripping activities. The backhand swing in tennis frequently exacerbates pain, as does gripping, repetitive wrist extension, and forearm pronation and supination. On physical examination, patients have tenderness on palpation of the lateral epicondyle. They also have positive Cozen and Mill tests. Entrapment of the posterior interosseous branch of the radial nerve mimics lateral epicondylitis, but the tenderness in this condition is 3 to 4 cm distal to the lateral epicondyle (at the level of the radial tunnel/supinator).¹³⁰

In the proper clinical context, radiographs do not need to be routinely performed. If there is diagnostic uncertainty, plain films assist in evaluating for elbow OA, a radial neck fracture (if the patient had a fall or other trauma), or calcific deposit in the tendons.²⁵

Treatment should emphasize proper work ergonomics, sport-ing technique, and corrective strengthening. Ensuring proper racquet grip size and string tension is essential. Eccentric strengthening of the wrist extensors appears to be the most effective exercise intervention.¹²⁶ Peritendinous corticosteroid injections are occasionally used, but their long-term efficacy is questionable.^{8,70} Additionally, this region is superficial, so complications such as subcutaneous fat atrophy and skin depigmentation are seen more often with steroid injections. Other injectables, such as autologous blood and PRP injections, have been used.^{31,84,87} Recent investigations have demonstrated superior efficacy of PRP and autologous blood injection compared with corticosteroid.^{11,38} Extracorporeal shock wave therapy has been reported to be successful in 48% to 73% of cases recalcitrant to other nonoperative

measures.⁹⁶ Recalcitrant cases can also be treated with surgical debridement of the degenerative tissue, with approximately 85% of surgically treated patients reporting good to excellent results and only 3% reporting no improvement.^{30,94}

Medial Epicondylitis (or Epicondylitis)

Medial epicondylitis, also known as golfer's elbow, has the same degenerative pathologic changes as those described in lateral epicondylitis.⁹⁸ The degenerative changes most commonly affect the pronator teres and flexor carpi radialis.⁹⁸ Risk factors include training errors, faulty equipment, and repetitive activities requiring wrist flexion and forearm pronation.

Patients typically report a gradual onset of pain over the medial epicondyle provoked by repetitive gripping, wrist flexion, and forearm pronation. The patient may report grip strength weakness. Physical examination demonstrates tenderness to palpation over the medial epicondyle, pain when a tight fist is made, and pain with resisted wrist flexion and forearm pronation.

Oblique radiographs of the medial epicondyle can reveal punctuate calcifications in the region of the flexor tendon origins.²⁵ It is also important to rule out degenerative changes in the posterior medial aspect of the olecranon because this condition can cause symptoms similar to those seen with medial epicondylitis.

Nonoperative treatment should include discontinuation of aggravating activities and use of analgesic medications, physical modalities, bracing (e.g., medial counterforce strap and neutral wrist splint) for pain control, along with correcting kinetic chain deficits and training errors. Eccentric strengthening of the flexor-pronator group is the standard of care. Corticosteroid injections have been used to treat this condition but can hasten degeneration; therefore clear goals regarding the role of corticosteroid injections should be established early in the treatment course. Autologous blood, PRP injections, and shock wave therapy may be useful adjuncts.^{96,125} Operative treatment is reserved for those who fail to improve with conservative measures.

Olecranon Bursitis

The olecranon bursa lies superficially over the olecranon process. Olecranon bursitis can be aseptic or septic. Aseptic bursitis is the result of acute hemorrhagic bursitis from trauma (e.g., fall or direct blow), repetitive microtrauma (e.g., leaning on the elbow), or gouty tophi.⁶⁶ The diagnosis is typically self-evident, with swelling and pain over the olecranon process. Traumatic bursitis will have a distinct injury, sudden onset, and signs of hemorrhage (warmth and erythema).

The treatment for traumatic and atraumatic bursitis is the same. PRICE principles are the foundation. Gentle compression with a compressive wrap or sleeve will reduce swelling. Avoidance of leaning on hard surfaces is crucial for eliminating continual irritation. For those who may have difficulty avoiding direct pressure, such as wheelchair-based patients, an elbow pad can be used for protection. Cryotherapy and NSAIDs may also be useful for pain control. Aspiration should be avoided unless the patient fails a dedicated trial of the above because the tissue is so superficial, and the risk of iatrogenic infection is high (up to 10%). Aspiration with concomitant steroid injections should be avoided because steroids further heighten the infection risk. If aspiration is performed, injection with a sclerosant (doxycycline) may reduce recurrence of swelling and have antimicrobial protective effects. Prophylactic antibiotics that cover for common skin flora may also be prescribed.⁴⁷ Surgical excision is reserved for patients who do not obtain relief with these measures.

The presence of inflammatory findings on examination should always raise the possibility of septic bursitis. Often, this can be

distinguished based on history; traumatic hemorrhagic bursitis will have a distinct mechanism of injury. In contrast, septic bursitis often lacks a mechanism of injury, and the patient frequently has significant comorbidities such as diabetes. Severe pain, fevers, and chills are also indicators of septic bursitis.

If septic bursitis is suspected, the bursa should be aspirated and sent for cell count, culture, and crystals. If testing confirms the presence of infection, intravenous antibiotics are the mainstay of treatment. Patients likely will need to be admitted through the local emergency room for this therapy.

Ulnar Collateral Ligament Injuries

The UCL is composed of an anterior, posterior, and transverse bundle. The anterior bundle provides the majority of stability, resisting valgus force when the elbow is flexed.¹²⁷ Injuries to the UCL of the elbow are a result of acute or repetitive valgus stress.

Possibly the most recognizable sporting scenario for UCL injuries is throwing in baseball. A detailed description of throwing mechanics is found in the sports medicine chapter of this text (Chapter 40). In brief, valgus stress is highest in the late cocking and acceleration phases.¹³¹ If that stress chronically overloads the UCL, degenerative changes occur. This results in chronic medial elbow pain during and after throwing. As degenerative changes accumulate, an acute-on-chronic injury may occur with a sudden pain and pop during a throw. In addition to medial elbow pain, the athlete may also experience ulnar nerve dysesthesias caused by traction and possible subluxation over the medial epicondyle. On examination, patients may demonstrate a 5-degree elbow flexion contracture, UCL tenderness, and pain during UCL stress testing (particularly the moving valgus stress test). Patients may also demonstrate ulnar subluxation with repeated flexion and extension.³⁵

Anteroposterior, lateral, and oblique radiographs of the elbow should be obtained. Calcification of the UCL can occur, and avulsions from the humeral or ulnar attachments should be ruled out.¹¹⁶ MRI can differentiate partial- vs. full-thickness injuries and can identify which bands are involved. Ultrasound has the added advantage that dynamic laxity can be assessed.

The treatment of UCL injuries is controversial, and no clear consensus exists for when operative intervention is needed, even in the setting of full-thickness injuries. Patient-specific factors such as symptom severity, timing with respect to competition, and level of play should be considered when designing the treatment plan. Even in the setting of high-grade injuries, nonoperative treatment may be favored because surgical outcomes can be uncertain. Studies cite that return to play can be up to 1 year after surgery and that over 40% of patients may experience chronic problems such as medial pain, recurrent injury, or kinesiophobia.³⁵ Therefore surgery should generally be reserved for patients who fail a comprehensive nonoperative treatment cycle. Many throwing athletes will respond to nonoperative treatment. Initial treatment should consist of rest, restoration of elbow ROM, strengthening of the dynamic medial stabilizers, and optimization of the kinetic chain function (including proper hip, spine, and shoulder mobility, and throwing technique). Once ROM, strength, and biomechanics are optimal, an interval throwing program should begin (Box 36.1). Data regarding PRP injections are conflicting, but these may be a reasonable step for injuries that do not respond to conservative care or to reduce pain when the patient is near or in the competitive season (so as to avoid a surgery that would end the season). If the patient fails to improve with the above, then surgical consultation should be sought.^{37,59}

• BOX 36.1 Interval Throwing Program

Athletes are allowed to progress from one phase to the next as soon as they are able to complete the phase without pain.

45-ft Phase

- Step 1: Warmup, twenty-five 45-ft throws $\times$ 2 with 15-min rest between sets
- Step 2: Warmup, twenty-five 45-ft throws $\times$ 3 with 15-min rest between sets

60-ft Phase

- Step 1: Warmup, twenty-five 60-ft throws $\times$ 2 with 15-min rest between sets
- Step 2: Warmup, twenty-five 60-ft throws $\times$ 3 with 15-min rest between sets

90-ft Phase

- Step 1: Warmup, twenty-five 90-ft throws $\times$ 2 with 15-min rest between sets
- Step 2: Warmup, twenty-five 90-ft throws $\times$ 3 with 15-min rest between sets

120-ft Phase

- Step 1: Warmup, twenty-five 120-ft throws $\times$ 2 with 15-min rest between sets
- Step 2: Warmup, twenty-five 120-ft throws $\times$ 3 with 15-min rest between sets

150-ft Phase

- Step 1: Warmup, twenty-five 150-ft throws $\times$ 2 with 15-min rest between sets
- Step 2: Warmup, twenty-five 150-ft throws $\times$ 3 with 15-min rest between sets

180-ft Phase

- Step 1: Warmup, twenty-five 180-ft throws $\times$ 2 with 15-min rest between sets
- Step 2: Warmup, twenty-five 180-ft throws $\times$ 3 with 15-min rest between sets
- Step 3: Athletes begin throwing off the mound or return to their respective positions

A distinct scenario is UCL injury in a nonthrowing athlete. Often these occur from a force valgus blow in combat sports or football. Because of the acute nature (as opposed to the chronic overuse observed in throwers), there is more significant local bleeding and therefore more healing; these injuries rarely require surgery. Most of these athletes will be able to return to sports within 2 to 4 weeks after resolution of symptoms and demonstration of ability to return safely to movement patterns involved in their sport. Initially, treatment should focus on acute pain control and restoration of ROM. As athletes demonstrate full strength and the ability to protect themselves, they may return to practice and competition under supervision. A hinged-elbow brace may be a useful tool to provide further protection while they are in the recovery phase. It also may reduce fear of reinjury as they return for the remainder of the season.¹¹³

Conditions of the Forearm, Wrist, and Hand

de Quervain Tenosynovitis

The first dorsal compartment of the wrist contains the abductor pollicis longus and extensor pollicis brevis tendons. These tendons run beneath a sheath over the dorsal aspect of the radial styloid process, along the inferior portion of the anatomic snuff box. Shear and repetitive microtrauma in this area can result in a tenosynovitis referred to as de Quervain tenosynovitis.¹¹⁰ This is the most common tendinopathy of the wrist and is frequently seen in patients who perform activities requiring gripping and/or repetitive use of the thumb.^{22,72}

Patients present with gradual onset of pain over the dorsal radial aspect of the wrist aggravated by activities such as racquet sports, golf, or fishing. Patients can also report a sensation of wrist

crepitus. Physical examination demonstrates mild edema localized to the dorsal radial wrist and tenderness to palpation over the first dorsal compartment. The Finkelstein test is pathognomonic for the diagnosis.¹¹⁰ It is performed by placing the patient's elbow in extension, with the forearm in neutral rotation and the wrist in radial deviation. The patient should be asked to place the thumb in the palm and grip the thumb with the fingers. The examiner should then passively deviate the wrist to the ulnar side. Pain over the first dorsal compartment is considered a positive test. Radiographs are typically not indicated for routine evaluation. Diagnostic ultrasound may show fluid in the tendon sheath.

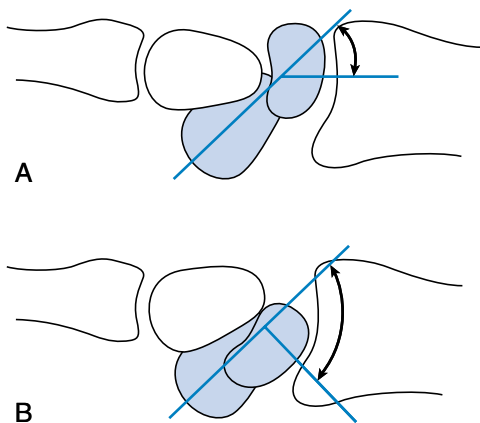
Treatment includes rest, analgesics, and a thumb spica splint. Pain refractory to this may benefit from a tendon sheath injection. First dorsal compartment peritendinous corticosteroid injection has been shown to reduce symptoms in 62% to 100% of cases.^{64,134}

Scapholunate Instability

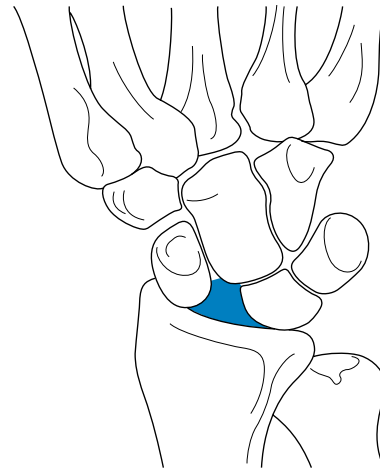
Scapholunate instability resulting from ligamentous injury is the most common ligamentous injury of the wrist.^{78,80} This injury occurs after a fall on a pronated, outstretched hand with the wrist in extension and ulnar deviation. After scapholunate ligament disruption, the scaphoid moves into a flexed position, whereas the lunate and triquetrum become extended. This pattern is referred to as dorsal intercalated segmental instability (DISI) pattern (Fig. 36.3).⁷⁰ This creates chronic instability of the carpus. If this injury is not diagnosed early and treated properly, the instability leads to progressive posttraumatic arthritis, which is termed *scapholunate advanced collapse (SLAC) wrist*.¹⁰⁹ Once a SLAC wrist has developed, there is no corrective treatment, and options are limited to palliative surgical treatment such as fusions and/or carpectomy. Therefore it is critical to identify and treat this injury early.

Scapholunate instability patients report a FOOSH injury with subsequent wrist edema, ecchymosis, and restricted ROM. They are tender over the scapholunate joint, particularly on the dorsum of the wrist. The patient typically has a positive result on the Watson test, in which shifting of the scaphoid relative to the lunate can be felt.⁷⁶

Radiographic evaluation should include anteroposterior, anteroposterior with a clenched fist, lateral, and oblique radiographs.¹⁰⁹ The presence of associated fractures should be assessed.



• **Fig. 36.3** (A) Normal scapholunate angle between 30 and 60 degrees. (B) Scapholunate angle greater than 60 degrees, indicating a dorsal intercalated segmental instability pattern suggestive of scapholunate dissociation.



• **Fig. 36.4** A gap between the scaphoid and lunate on anteroposterior radiographs indicates scapholunate dissociation.

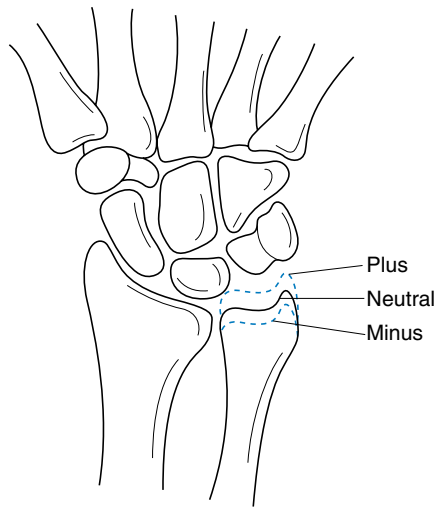
The lateral radiograph can reveal the presence of a DISI pattern injury (see Fig. 36.3). This is characterized by a scapholunate angle of more than 60 degrees.¹⁰⁹ A gap of more than 3 mm between the scaphoid and lunate on anteroposterior radiographs is also diagnostic for scapholunate instability (Fig. 36.4).¹⁰⁹ The clenched fist view exaggerates this finding. Radiographic evaluation should include comparison studies with the opposite wrist.

If standard radiographs are not diagnostic, an MRI should be obtained.¹⁰⁹ Wrist arthroscopy has become the gold standard for diagnosing this entity. Acute scapholunate ligament injuries should be treated surgically. Chronic scapholunate injuries are more difficult to treat but also require surgical intervention. Partial wrist arthrodesis can improve chronic scapholunate instability, and a proximal row carpectomy frequently is used to treat advanced scapholunate collapse.¹⁰⁹

Triangular Fibrocartilage Complex Injuries

The triangular fibrocartilage complex (TFCC) is composed of an avascular central articular disc and vascular dorsal and palmar radioulnar ligaments.¹⁰⁹ The TFCC is the primary stabilizer of the distal radioulnar joint and can be injured in an acute traumatic event, such as falling on an outstretched hand, or through repetitive microtrauma, such as that seen in gymnastics. Axial loads to the wrist result in 18% of the load transmitted through the TFCC and the remaining 82% through the radiocarpal joint.¹⁰¹ A positive ulnar variance (longer ulna than radius at the level of the wrist) results in an increase in the load-bearing function of the TFCC, which results in a higher incidence of TFCC injuries (Fig. 36.5).¹⁰¹

Patients who sustain a TFCC injury may report either a gradual onset or a single traumatic event. Traumatic tears occur more frequently in young athletes, whereas degenerative tears are more common in older patients. Patients with acute injuries often report an axial load to the wrist associated with rotational stress.¹⁰⁹ The patient can also report wrist catching. The physical examination can show tenderness to palpation in the hollow between the flexor carpi ulnaris tendon and the extensor carpi ulnaris (ECU) tendon, just distal to the ulnar styloid process.¹⁰⁹ This should be differentiated from ECU tendinitis, which is more common in racquet sports and may have reproducible subluxation on pronation/supination.



• **Fig. 36.5** Neutral, ulnar minus, and ulnar plus variants.

Radiographic evaluation may demonstrate an ulnar-plus variant. Wrist arthrography, MRI, and MR arthrography can provide more specific information regarding TFCC pathology.¹⁰⁹

Initial treatment should consist of immobilization with a wrist splint, rest, pain control measures (NSAIDs or a steroid injection), and optimization of mechanics (especially for athletes who preferentially load their wrists). If patients fail to respond adequately, evaluation for surgery (consisting of repair vs. debridement, with or without ulnar shortening) should occur.¹⁶

First Metacarpophalangeal Joint Ulnar Collateral Ligament Sprain

Radially directed forces across the first metacarpophalangeal (MCP) joint can result in UCL injury. This injury occurs after injuries such as a fall or impact in which the thumb is forced into radial deviation.¹¹⁰ A fall in downhill skiing is a common cause when the gloved thumb is forced into radial deviation by contact with the ground. UCL injuries to the first MCP joint can be categorized into a three-grade severity scale of ligament sprains (see Table 36.2).

When a grade 3 UCL sprain of the first MCP joint occurs as a result of avulsion of the distal end of the ligament from the base of the first proximal phalanx, there is the possibility of interposition of the adductor pollicis aponeurosis between the base of the first proximal phalanx and the ruptured end of the UCL.¹²³ This is called a *Stener lesion* and prevents healing of the first MCP UCL injury, leading to chronic joint pain and instability.

Patients who sustain first MCP joint UCL injuries report a radially directed force across the first MCP joint. They may

report an associated “pop” and a feeling of instability in the joint. Physical examination reveals tenderness to palpation over the UCL. If a Stener lesion is present, a palpable mass on the ulnar side of the first MCP joint may be present, representing the avulsed UCL.² Patients also demonstrate swelling of the first MCP, ecchymosis, and increased laxity during radial stress. The stress examination should be performed with the joint in both full extension and 30 degrees of flexion. A complete tear is indicated by an angular difference between the injured and uninjured first MCP joint during stress examination of greater than 15 to 30 degrees.^{54,100} A lack of an endpoint during stress examination also suggests complete UCL disruption.

Radiologic evaluation should include anteroposterior, lateral, and oblique radiographs to detect the presence of fractures or joint subluxation. MRI and diagnostic ultrasound can visualize the soft tissue injuries.

Treatment of partial tears involves modalities, analgesics, and immobilization for 6 weeks.³⁹ Patients who participate in contact sports should continue to wear a thumb spica splint with a soft covering during competition for the remainder of the season. Local taping for stability during activity can be used after the period of splinting has been completed. Gentle progressive ROM exercises should begin after cast immobilization by removing the splint twice daily, and activity should progress as tolerated.

Early surgical repair has been advocated for complete ruptures of the UCL, particularly if a Stener lesion is present.^{34,69,81,99,119} Surgery is also indicated for individuals with an avulsion fracture of the base of the proximal phalanx with angulation and displacement greater than 3 mm or with chronic recurrent instability.⁸⁵ In-season athletes may play in a soft cast with a plan for postseason surgical correction. This decision should be made in conjunction with the surgical team and with the patient fully understanding the risks and benefits of delayed surgery.

Summary

There are many common upper limb injuries and syndromes. Although most can be treated nonoperatively, some require surgical intervention. Accurate diagnosis and initiation of appropriate treatment will enhance recovery, and identification of flaws in the movement chain can help to optimize rehabilitation and prevent recurrent injury.

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TABLE 36.2 Ligamentous Injury Grading Scale

Grade	Signs
1	Tenderness to palpation without joint laxity
2	Tenderness to palpation with joint laxity but a good endpoint
3	Tenderness to palpation with significant joint laxity and no endpoint

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